

Chapter 4, Entropy rates of a stochastic process

Recall AEP

X_i i.i.d with probability dist. $p(x)$.
stationary ergodic process.

$-\frac{1}{n} \log p(X_1, X_2, \dots, X_n) \rightarrow H(X)$ in probability.

$\rightarrow p(X_1, X_2, \dots, X_n) \approx 2^{-nH(X)}$ "entropy rate"

$\rightarrow$ Typical set $A_\epsilon^{(n)} := \{ (x_1, x_2, \dots, x_n) \in \mathcal{X}^n \mid 2^{-n(H+\epsilon)} \leq p(x_1, x_2, \dots, x_n) \leq 2^{-n(H-\epsilon)} \}$

$\rightarrow$ Data compression.

Q) How many bits are needed to describe X_i : dependent?

Sec 16.8

1. entropy rate, Markov chain.
2. entropy rate of stationary Markov chain
3. " functions of Markov chain
4. 2nd law of thermodynamics

§. Markov chain, entropy rate.

• joint p.d.f $p(x_1, x_2, \dots, x_n) = \Pr(X_1=x_1, X_2=x_2, \dots, X_n=x_n) \prod_{i=1}^{n-1} p(x_{i+1}|x_i)$ $\forall n$.

Markov chain
discrete

discrete stochastic proc. $\{X_n | n \in \mathbb{N}\}$ s.t.

$$\Pr(X_{n+1}=j | X_n=i, X_{n-1}, \dots, X_1) = \Pr(X_{n+1}=j | X_n=i).$$

$$\rightarrow p(x_1, \dots, x_n) = p(x_1) p(x_2|x_1) p(x_3|x_2) \dots p(x_n|x_{n-1})$$

stationary process. $\Pr(X_1=x_1, \dots, X_n=x_n) = \Pr(X_{1+\ell}=x_1, X_{2+\ell}=x_2, \dots, X_{n+\ell}=x_n) \forall \ell, \forall n$.

time-invariant Markov chain s.t.

$$\Pr(X_{n+1}=j | X_n=i) = \Pr(X_2=j | X_1=i) =: p_{ij}$$

→ For states $1, 2, \dots, m$,

Initial dist. & transition matrix determine such process.

$$P = \begin{matrix} & \begin{matrix} 1 & 2 & \dots & j & \dots & m \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ \vdots \\ i \\ \vdots \\ m \end{matrix} & \begin{pmatrix} & & & & \\ & & & & \\ & & & & \\ \hline & & & p_{ij} & \\ \hline & & & & \end{pmatrix} \end{matrix}$$

$$\Pr(X_{n+1} = j) = \sum_{k=1}^m \Pr(X_n = k) p_{kj}$$

irreducible

Markov chain s.t.

$\forall i, j \in \{1, 2, \dots, m\}, \exists n \in \mathbb{N}$ s.t.

$$\Pr(X_n = j | X_1 = i) > 0$$

aperiodic

irreducible Markov chain

let d : largest integer s.t.

$$\Pr(X_{n+d} = i | X_1 = i) > 0 \Rightarrow n \mid d \text{ for all } i$$

Then, $d=1$.

Stationary distribution.

distribution $\mu = (\mu_1, \mu_2, \dots, \mu_m)$ s.t.

$$\Pr(X_n = i) = \mu_i, \quad \forall i = 1, 2, \dots, m$$

$$\Rightarrow \Pr(X_{n+1} = i) = \mu_i, \quad \forall i, \forall n$$

* μ is stationary distribution $\Leftrightarrow \mu = \mu P$

* finite-state Markov chain is irreducible & aperiodic $\Rightarrow$

stationary dist. is unique, $\mu_i = \lim_{n \rightarrow \infty} \Pr(X_n = i)$ for any initial dist.

entropy rate.

1. Given $\{X_i\}$: stochastic process,

$$H(X) := \lim_{n \rightarrow \infty} \frac{1}{n} H(X_1, X_2, \dots, X_n).$$

2. Given $\{X_i\}$: stochastic process,

$$H'(X) := \lim_{n \rightarrow \infty} H(X_n | X_{n-1}, \dots, X_1).$$

ex). typewriter. m symbols, n letters

$$\begin{aligned} \Rightarrow H(X_1, X_2, \dots, X_n) &= - \sum_{x_1, x_2, \dots, x_n} p(x_1, \dots, x_n) \log p(x_1, \dots, x_n) \\ &= - \sum \frac{1}{m^n} \log \frac{1}{m^n} \\ &= \log m^n \end{aligned}$$

$$H(X) = \lim_{n \rightarrow \infty} \frac{\log m^n}{n} = \log m.$$

$$\begin{aligned} \Rightarrow H(X_n | X_{n-1}, \dots, X_1) &= - \sum_{x_1, \dots, x_n} p(x_1, \dots, x_n) \log p(x_n | x_{n-1}, \dots, x_1) \\ &= - \sum \frac{1}{m^n} \log \frac{1}{m} \\ &= \log m \end{aligned}$$

$$H'(X) = \log m.$$

thm 4.2.1 $\{X_i\}$: stationary stochastic process $\Rightarrow H(X) = H'(X).$

$$\begin{aligned} \text{pf). } H(X_{n+1} | X_n, X_{n-1}, \dots, X_1) &\leq H(X_{n+1} | X_n, \dots, X_2) \\ &= H(X_n | X_{n-1}, \dots, X_1). \end{aligned}$$

$H(X_n | X_{n-1}, \dots, X_1)$ is decreasing, nonnegative, so has a limit $H'(X).$

By chain rule, note that.

$$b_n = \frac{1}{n} H(X_1, X_2, \dots, X_n) = \frac{1}{n} \sum_{i=1}^n H(X_i | X_{i-1}, \dots, X_1) \quad a_i$$

Claim). $a_n \rightarrow a$, $b_n = \frac{1}{n} \sum_{i=1}^n a_i$, then $b_n \rightarrow a$.

Pf of Claim). $\forall \epsilon > 0$, $\exists N$ s.t. $n \geq N$ implies $|a_n - a| \leq \epsilon$.

$$\begin{aligned} |b_n - a| &= \left| \frac{1}{n} \sum_{i=1}^n (a_i - a) \right| \\ &\leq \frac{1}{n} \sum_{i=1}^n |a_i - a| \end{aligned}$$

$$\leq \frac{1}{n} \sum_{i=1}^N |a_i - a| + \frac{1}{n} (n-N) \varepsilon$$

$$< 2\varepsilon \quad \text{for sufficiently large } n.$$

$$\lim_{n \rightarrow \infty} \frac{1}{n} H(X_1, X_2, \dots, X_n) = H(X) \text{ exists and } H(X) = H'(X).$$

p.

§. Entropy rate of stationary Markov chain.

Suppose $\{X_i\}$: stationary Markov chain.

$$\begin{aligned} \Rightarrow H(X) &= H'(X). \\ &= \lim_{n \rightarrow \infty} H(X_n | X_{n-1}, \dots, X_1). \\ &= \lim_{n \rightarrow \infty} H(X_n | X_{n-1}) \\ &= \lim_{n \rightarrow \infty} H(X_2 | X_1) \\ &= H(X_2 | X_1) \end{aligned}$$

let M : stationary dist. of $\{X_i\}$.

Assume $x_1 \sim M$

$$\begin{aligned} \Rightarrow \underline{H(X)} &= H(X_2 | X_1) \\ &= - \sum_{x_1, x_2} p(x_1, x_2) \log p(x_2 | x_1). \\ &= - \sum_{x_1, x_2} M_{x_1} P_{x_1, x_2} \log P_{x_1, x_2}. \\ &= \underline{- \sum_{i,j} M_i P_{ij} \log P_{ij}} \quad (*) \end{aligned}$$

* Same applies to irred & aperiodic Markov chain.

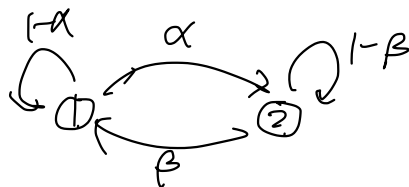
↓
 $\exists!$ stationary dist.
 ↓

Even if $X_1 \notin M$, (*) holds

ex). Consider a Markov chain st

State = $\{1, 2\}$.

$$P = \begin{matrix} & \begin{matrix} 1 & 2 \end{matrix} \\ \begin{matrix} 1 \\ 2 \end{matrix} & \begin{pmatrix} 1-\alpha & \alpha \\ \beta & 1-\beta \end{pmatrix} \end{matrix}$$



$\Rightarrow$ let μ : stationary dist.

$$\mu = \mu P$$

$$(\mu_1 \ \mu_2) = (\mu_1 \ \mu_2) \begin{pmatrix} 1-\alpha & \alpha \\ \beta & 1-\beta \end{pmatrix}.$$

$$\mu_1 = \mu_1(1-\alpha) + \mu_2\beta.$$

$$\begin{cases} \mu_1\alpha = \mu_2\beta \\ \mu_1 + \mu_2 = 1 \end{cases}$$

$$\mu_1 = \frac{\beta}{\alpha+\beta}, \quad \mu_2 = \frac{\alpha}{\alpha+\beta}.$$

Suppose $X_1 \sim \mu$, so that $\{X_i\}$ is stationary.

$$\Rightarrow H(X) = H(X_2|X_1)$$

$$= - \sum_{i,j} \mu_i P_{ij} \log P_{ij}$$

$$= - \mu_1 \sum_j P_{1j} \log P_{1j} - \mu_2 \sum_j P_{2j} \log P_{2j}$$

$$= \frac{\beta}{\alpha+\beta} H(\alpha) + \frac{\alpha}{\alpha+\beta} \cdot H(\beta).$$

$$\left(H(p) := -p \log p - (1-p) \log (1-p) \right).$$

§ Entropy rate of functions of Markov chains.

Suppose $\{X_i\}$: stationary Markov chain

$$Y_i = \phi(X_i)$$

$\Rightarrow \{Y_i\}$: not necessarily Markov chain,
but is stationary proc, so $H(Y)$ is well defined.

Q) What is $H(Y) = \lim H(Y_n | Y_{n-1}, \dots, Y_1)$?

Good: we present the bounds of $H(Y_n | Y_{n-1}, \dots, Y_1)$.

* $H(Y_n | Y_{n-1}, \dots, Y_1)$: decreasing, nonnegative, so it converges.

lem 4.5.1.

$$H(Y_n | Y_{n-1}, \dots, Y_2, X_1) \leq H(Y).$$

$$\begin{aligned} \text{pf). } H(Y_n | Y_{n-1}, \dots, Y_2, X_1) &= H(Y_n | Y_{n-1}, \dots, Y_2, Y_1, X_1) \\ &= H(Y_n | Y_{n-1}, \dots, Y_2, Y_1, X_1, X_0, X_{-1}, \dots, X_{-k}) \\ &= H(Y_n | Y_{n-1}, \dots, Y_2, Y_1, Y_0, \dots, Y_{-k}, X_1, X_0, \dots, X_{-k}) \\ &\leq H(Y_n | Y_{n-1}, \dots, Y_1, Y_0, \dots, Y_{-k}). \end{aligned}$$

$$\begin{aligned} &\overline{\uparrow} H(Y_{n+k+1} | Y_{n+k}, \dots, Y_1) \\ &\quad \downarrow \\ &\quad \begin{array}{l} Y_i: \text{stationary.} \\ H(Y). \end{array} \end{aligned}$$

lem 4.5.2.

$$H(Y_n | Y_{n-1}, \dots, Y_1) - H(Y_n | Y_{n-1}, \dots, Y_1, X_1) \rightarrow 0.$$

$$\text{pf). } H(Y_n | Y_{n-1}, \dots, Y_1) - H(Y_n | Y_{n-1}, \dots, Y_1, X_1) = I(X_1; Y_n | Y_{n-1}, \dots, Y_1).$$

(Recall $I(X:Y|Z) = I(Y:X|Z) = H(Y|Z) - H(Y|X,Z)$).

Note by chainrule of mutual information,

$$I(X:Y_1, \dots, Y_n) = \sum_{i=1}^n I(X_1:Y_i | Y_{i-1}, \dots, Y_1).$$

$\Rightarrow$, $H(X_i) \geq I(X_i:Y_1, \dots, Y_n)$, and $I(X_i:Y_1, \dots, Y_n)$ is increasing with n , hence $\lim_{n \rightarrow \infty} I(X_i:Y_1, \dots, Y_n)$ exists.

$$\begin{aligned} & (\because I(X_i:Y_1, \dots, Y_n) - I(X_i:Y_1, \dots, Y_{n-1}) \\ & = I(X_i:Y_n | Y_1, \dots, Y_{n-1}) \geq 0) \end{aligned}$$

$$\begin{aligned} \Rightarrow \lim_{n \rightarrow \infty} I(X_i:Y_1, \dots, Y_n) &= \lim_{n \rightarrow \infty} \sum_{i=1}^n I(X_i:Y_i | Y_{i-1}, \dots, Y_1) \\ &= \sum_{i=1}^{\infty} I(X_i:Y_i | Y_{i-1}, \dots, Y_1), \text{ converges.} \end{aligned}$$

$$\Rightarrow \lim_{n \rightarrow \infty} I(X_i:Y_n | Y_1, \dots, Y_{n-1}) = 0.$$

thm 4.5.1.

$\{X_i\}$: stationary Markov chain.

$$Y_i = \phi(X_i).$$

$$\Rightarrow 1. \quad H(Y_n | Y_{n-1}, \dots, Y_1, X_1) \leq H(Y) \leq H(Y_n | Y_{n-1}, \dots, Y_1).$$

$$2. \quad \lim_{n \rightarrow \infty} H(Y_n | Y_{n-1}, \dots, Y_1, X_1) = H(Y) = \lim_{n \rightarrow \infty} H(Y_n | Y_{n-1}, \dots, Y_1).$$

* Hidden Markov model. (HMM).

Y_i is drawn according to $p(y_i | x_i)$, so that,

$$p(x_1, x_2, \dots, x_n, y_1, \dots, y_n) = p(x_1) \prod_{i=1}^{n-1} p(x_{i+1} | x_i) \prod_{i=1}^n p(y_i | x_i)$$

§. Second law of thermodynamics.

* Second law of thermodynamics: Entropy of an isolated sys. is nondecreasing.

1. Relative entropy $D(M_n \| M'_n)$ decreases with n .

let M_n, M'_n : probability dist. of Markov chain at time n
with joint mass ftn p, q resp.

let r : transition probability ftn.

$$\Rightarrow p(x_n, x_{n+1}) = p(x_n) r(x_{n+1} | x_n)$$

$$q(x_n, x_{n+1}) = q(x_n) r(x_{n+1} | x_n)$$

$$\begin{matrix} 0 \\ \| \\ D(r(x_{n+1} | x_n) \| r(x_{n+1} | x_n)) \end{matrix}$$

$\Rightarrow$ By chain rule,

$$\begin{aligned} D(p(x_n, x_{n+1}) \| q(x_n, x_{n+1})) &= D(p(x_n) \| q(x_n)) + \boxed{D(p(x_{n+1} | x_n) \| q(x_{n+1} | x_n))} \\ &= D(p(x_{n+1}) \| q(x_{n+1})) + \underbrace{D(p(x_n | x_{n+1}) \| q(x_n | x_{n+1}))}_{\geq 0} \end{aligned}$$

$$\Rightarrow D(p(x_n) \| q(x_n)) \geq D(p(x_{n+1}) \| q(x_{n+1})).$$

$$\text{i.e. } D(M_n \| M'_n) \geq D(M_{n+1} \| M'_{n+1}).$$

2. Relative entropy $D(M_n \| M)$, where M is stationary dist., decreases with n .

let M : stationary dist., $M'_n = M$.
then $M'_{n+1} = M$.

$$\Rightarrow \text{By 1. } D(M_n \| M) \geq D(M_{n+1} \| M).$$

$\Rightarrow D(M_n \| M)$ is nonnegative, decreasing, so has a limit.

If M is unique, then $\lim_{n \rightarrow \infty} D(M_n \| M) = 0$

3. Entropy increases if the stationary dist. is uniform.

Suppose M : stationary dist, with prob mass fn u .

$$\begin{aligned}\Rightarrow D(M_n \| M) &= \sum_{x_n \in X} p(x_n) \log \frac{p(x_n)}{u(x_n)} \\ &= \sum_{x_n \in X} p(x_n) \log \frac{1}{u(x_n)} = H(X_n) \\ &= \log |X| = H(X_n).\end{aligned}$$

By 2, $H(X_n) \leq H(X_{n+1})$.

ex). Suppose M : nonuniform d.
 M' : uniform d.

$$X_1 \sim M'$$

$$\Rightarrow H(X_n) = \log |X| > H(X_1), \text{ for large } n.$$

$\therefore$ Entropy does not increase

Remark.

Uniform dist is a stationary dist of Markov chain

$\Leftrightarrow P$ is doubly stochastic.

$$\left(\begin{array}{l} \text{i.e. } \sum_j P_{ij} = 1, \forall j \\ \sum_i P_{ij} = 1, \forall i \end{array} \right)$$

4. Conditional entropy $H(X_n | X_1)$ increases with n for stationary Markov process.

$\{X_i\}$: stationary Markov process.

$$\Rightarrow H(X_n) = - \sum_{x_n \in \mathcal{X}} p(x_n) \log p(x_n). \quad \text{is constant with } n.$$

$$\Rightarrow H(X_n | X_1) \geq H(X_n | X_1, X_2).$$

$$= H(X_n | X_2).$$

$$= H(X_{n-1} | X_1).$$

' $H(X_n | X_1)$ increases with n .